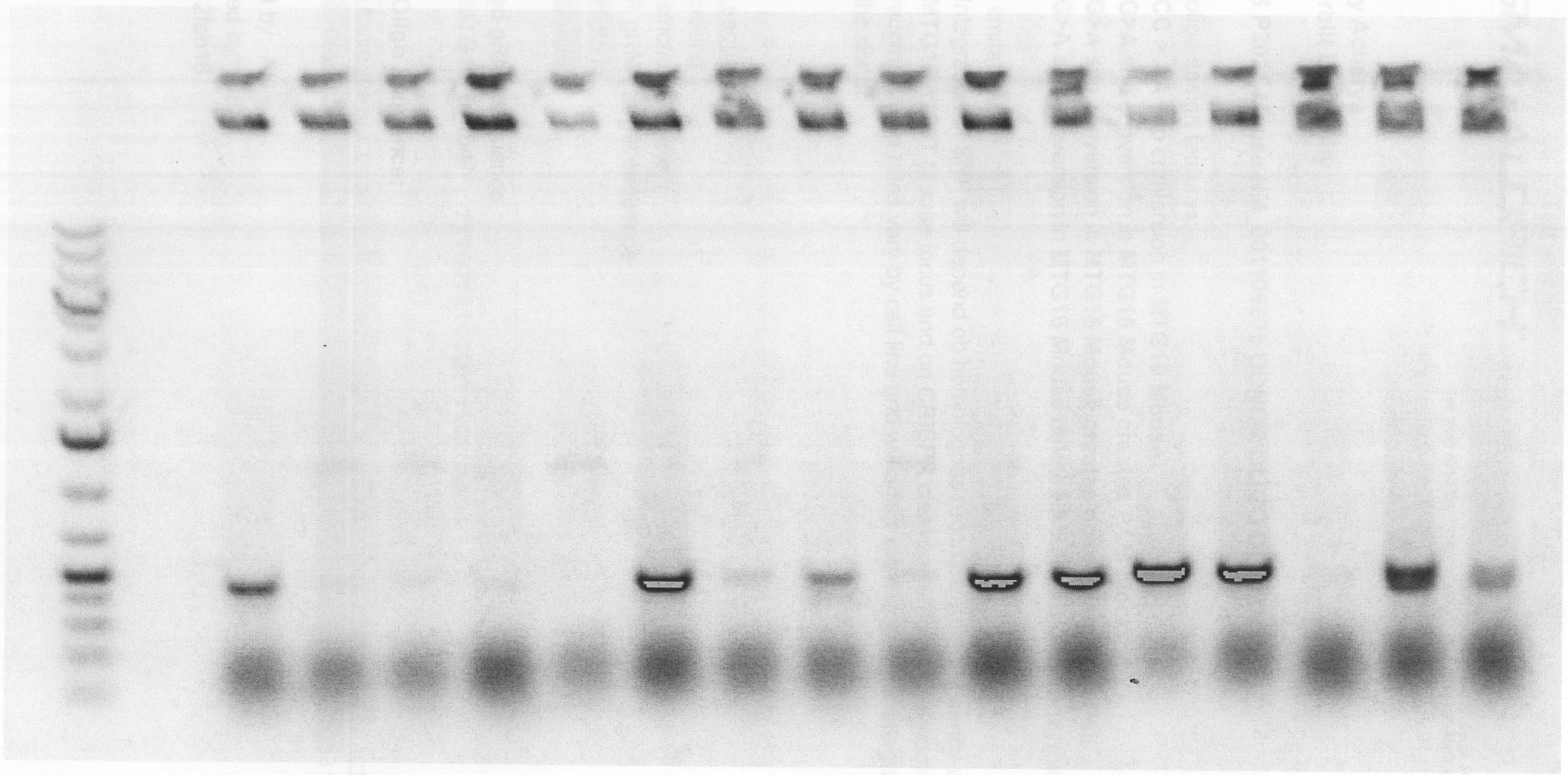
PCR screening of MTG16-P209T mutants



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3050 Spruce Street, St. Louis, MO 63103 USA
Tel: (800) 521-8956 (314) 771-5765 Fax: (800) 325-5052 (314) 771-5757

Pankay Acharya
Vanderbilt University

MTG16 P209T Mouse NM_001109873 DONOR OLIGO DESIGN

Donor oligo design features:

1) **CC > AA** snp conversion in **MTG16 Mouse**
2) **G>A** snp conversion in **MTG16 Mouse creates PAM blocking**
3) **G>A** snp conversion in **MTG16 Mouse creates novel restriction PsiI**
4) **G>A** snp conversion in **MTG16 Mouse deletes AcuI restriction enzyme**

Design limitations:

1) Integration of SNP will depend on homologous recombination frequency at the **MTG16 P209T Mouse** locus and on CRISPR cleavage efficiency. Homologous recombination rates vary by cell line and will affect the efficiency of integration of the donor.

KEY:
SNP
Coding sequence
PAM blocking
PsiI Restriction enzyme
CRISPR gRNA target sequence
Left Homology
Right Homology

Validated gRNA sequence:
Mouse-MTG16_0_154 TTTGTTATCCCTTTTCTGAAGG

Donor Oligo Sequence:
5′gtttcatgccaagctccaggaagccaccaactttccactgaggccgtttg*ttat*AActttttc
tgaaAgtaagcgaaagcagcaccttccaggcaacagggatggtgtacacaaaggccct3′

Created by: AH
Reviewed by: CB
Date: 12feb16

MJG16-P209T

```
P gccagatctggaacctcctcctcctcctcccccctcctcctcctcctcccccctcctcccc
3'  cggtctagaccttggaggaggaggaggagggggaggaggaggaggagggggaggagggg        65

aaggtgtgggtggggcatctgggtcagtaatgcatgggaacctttagatggggcggtcggtctct
ttccacacccaccccgtagacccagtcattacgtacccttggaaatctaccccgccagccagaga   130

ctgcagAACTCAACTCTGACGATCGAAGAGTTTCATGCCAAGCTCCAGGAAGCCACCAACTTTCC
gacgtcTTGAGTTGAGACTGCTAGCTTCTCAAAGTACGGTTCGAGGTCCTTCGGTGGTTGAAAGG   195
      1         5             10              15             20
      N  S  T  L  T  I  E  E  F  H  A  K  L  Q  E  A  T  N  F  P
      Exon5 >

ACTGAGGCCGTTTGTTATCCCTTTTCTGAAGgtaagcgaaagcagcaccttccaggcaacaggga
TGACTCCGGCAAACAATAGGGAAAAGACTTCcattcgctttcgtcgtggaaggtccgttgtccct   260
            25              30
 L  R  P  F  V  I  P  F  L  K  V  S  E  S  S  T  F  Q  A  T  G
 Exon5 > ------------- (in frame with Exon5) ------------->
                   P209 > F210 >
                           |
                          F210

tggtgtacacaaaggccctgggaaagaagtgtatacaaagctgctttgtaatggctggaatccag
accacatgtgtttccgggaccctttcttcacatatgtttcgacgaaacattaccgaccttaggtc   325
M  V  Y  T  K  A  L  G  K  K  C  I  Q  S  C  F  V  M  A  G  I  Q
----------------------- (in frame with Exon5) ------------------->

tgtgggaaatgggtgtgcagaggatgaagaacaggtgatggacaggcaggtgtgtgcctggaccg
acacccttacccacacgtctcctacttcttgtccactacctgtccgtccacacacggacctggc   390
 C  G  K  W  V  C  R  G  *
------- (in frame with Exon5) ------->

cagggtgacagaagatgctgtgccaaatgcagactggggcctcacaaacgacggtgatgccgtac
gtcccactgtcttctacgacacggtttacgtctgaccccggagtgtttgctgccactacggcatg   455

taggtccctttggcagagcaaccag        3'
                                     480
atccagggaaaccgtctcgttggtc P      5'
```